

Objectives

A. Back-end Modules

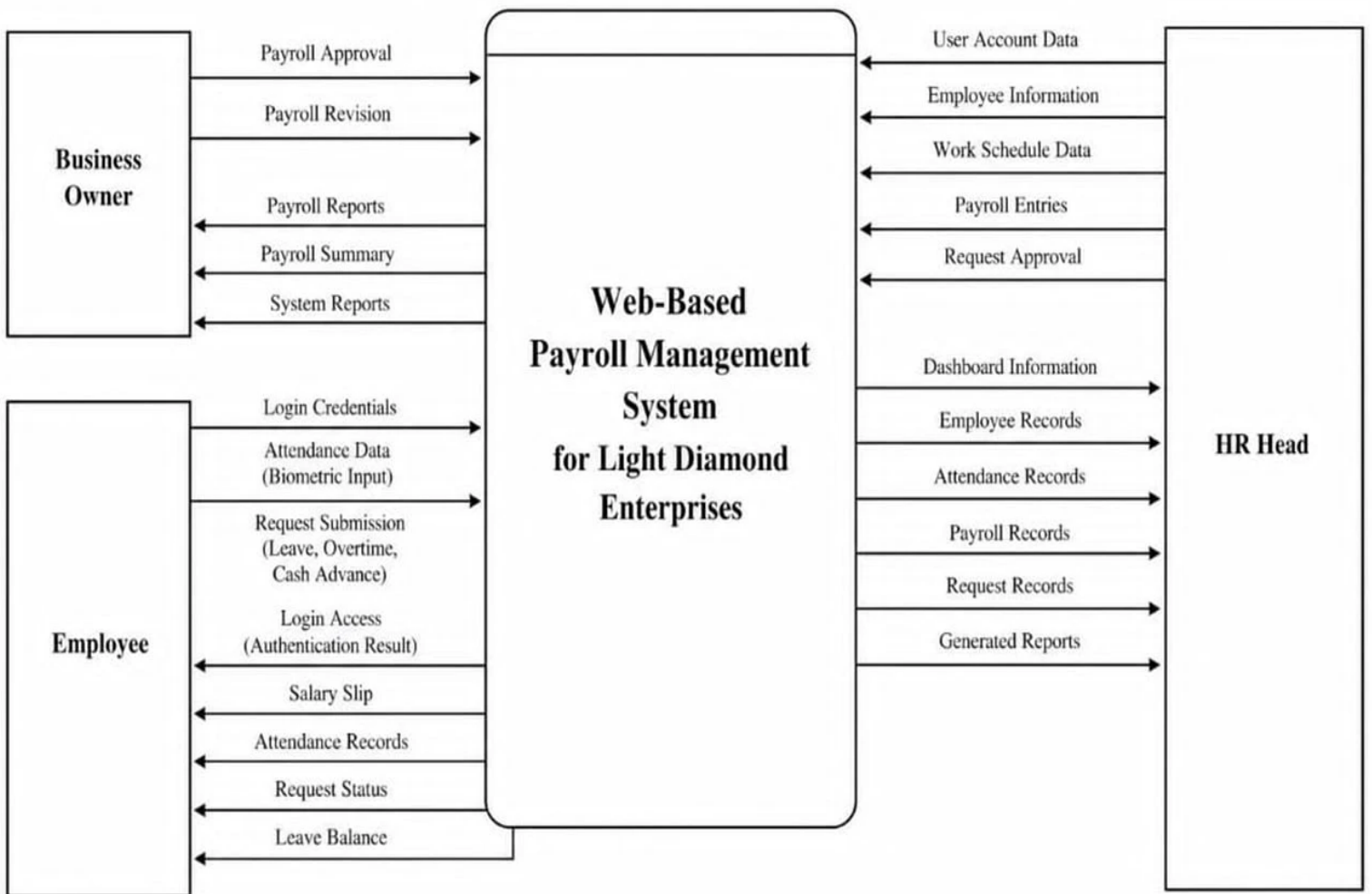
- 1) Login and Authentication
- 2) Dashboard → role-based summary
- 3) User Management → regulates access and user accounts
- 4) Employee Management → maintains employee records/information
- 5) Attendance Management → computes attendance data w/ direct biometric integration
- 6) Work Schedule → monitoring schedule and missing entries
- 7) Request Management → request status and records
- 8) Payroll Management → automates net salary calculation
- 9) Manage Salary → manage salary adjustments
- 10) Benefits and Deductions → calculates gov't contributions
- 11) Reports Management → payroll summaries, attendance records, request reports, Contribution reports, and 13th month pay summaries.

B. Front-End Modules

- 1) Employee Login → login and recovery
- 2) Attendance Viewing → monitoring of work performance
- 3) Request → submit and track
- 4) Payroll Viewing → view and download

Problems Identified:

- 1) Unprotected Excel payroll file (unauthorized edits → data inaccuracies)
- 2) Manual verification of attendance (time consuming consolidation)
- 3) Payroll computation using Excel formulas (inaccurate salary calculations)
- 4) Cash advance and leave (not properly integrated into payroll)
- 5) Historical payroll records (difficulty in retrieving)

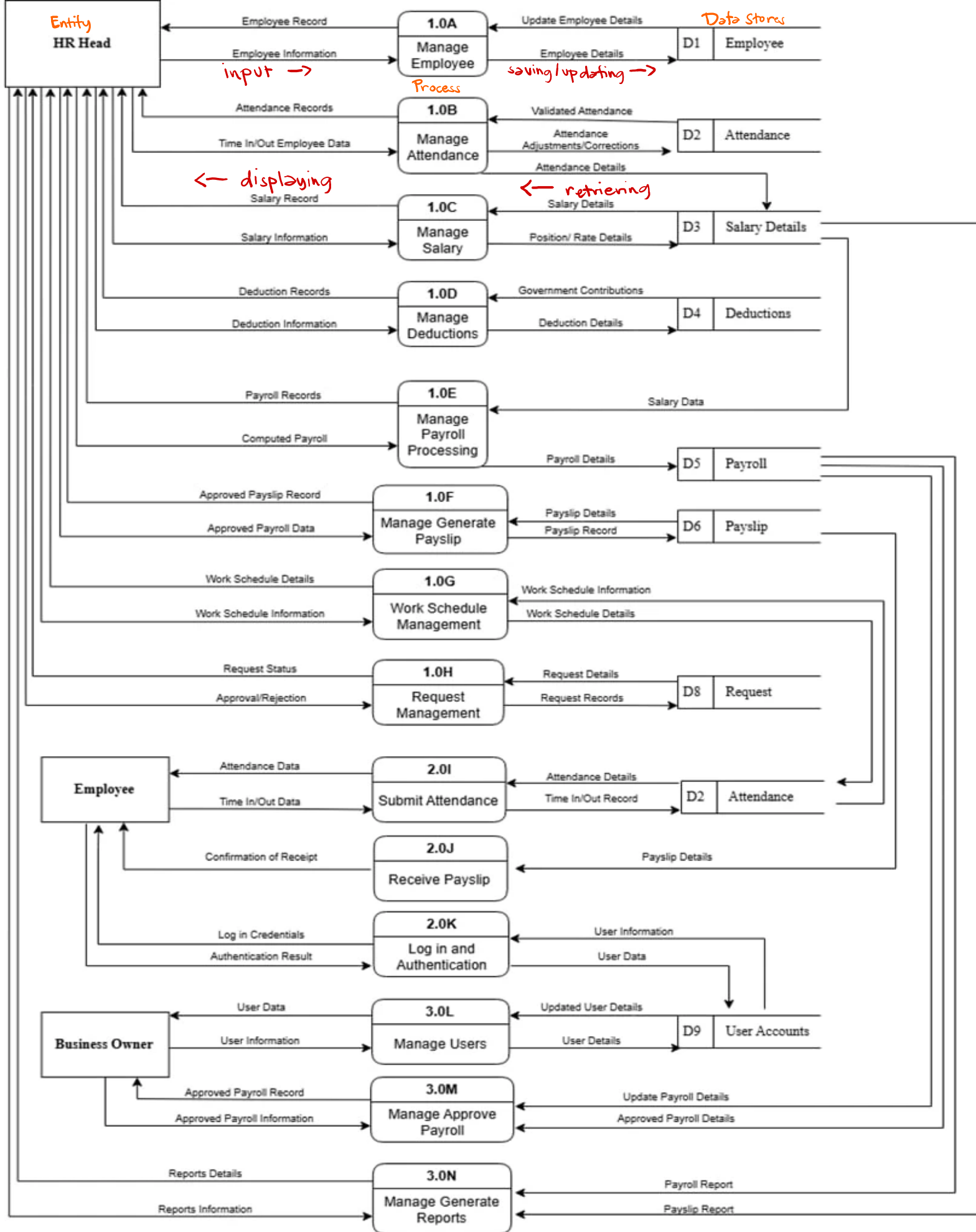


Context Diagram → focused on Entities and data flow

DFD (Level 0) → focused on Processes and databases

Technical Specifications

a)



LEGENDS

Process A: Manage Employee
 Process B: Manage Attendance
 Process C: Manage Salary
 Process D: Manage Deductions
 Process E: Manage Payroll Processing
 Process F: Manage Generate Payslip
 Process G: Work Schedule Management

Process H: Request Management
 Process I: Submit Attendance
 Process J: View Receive Payslip
 Process K: Log in and Authentication
 Process L: Manage Users
 Process M: Manage Approve Payroll
 Process N: Manage Generate Reports

PAYROLL CALCULATIONS

Fixed Variables

a) Daily Rate $\rightarrow 460 \text{ php}$

b) Work Hours $\rightarrow 8 \text{ hrs/day}$

c) Hourly Rate $\rightarrow 460/8 = 57.50 \text{ php}$

d) Late Deduction $\rightarrow 1 \text{ php/minute}$

e) CA Repayment $\rightarrow 500 \text{ php per week}$

f) Monthly Basic Salary $\rightarrow 11,998.33 \text{ php}$

Estimated Equivalent Monthly Rate (EEMR)

$\rightarrow$ constant multiplier to find average monthly salary.

* $365 \text{ days} - 52 \text{ rest days} = 313 \text{ days (Standard Factor)}$

DOLE Formula:

$\text{EEMR} = (\text{Daily Rate} \times \text{Total Working Days in a Year}) \div 12 \text{ months}$

$= (460 \times 313 \text{ days}) = 143,980 \text{ (Annual Salary)}$

$= 143,980 \div 12 \text{ months} = 11,998.33 \text{ php}$

Note: Monthly Basic Salary is used to accurately determine SSS bracket & PhilHealth contribution

Monthly Contributions (deducted every last Friday of the month)

a) SSS ($\text{P}12,000$ MSC Bracket) $\rightarrow 1,680 \text{ php (14\%)}$

* Employee Share (4.5%) $\rightarrow 540 \text{ php}$

* Employer Share (9.5%) $\rightarrow 1,140 \text{ php}$

b) PhilHealth (4% of 11,998.33) $\rightarrow 479.93 \text{ php}$

* Employee Share (50%) $\rightarrow 239.97 \text{ php}$

* Employer Share (50%) $\rightarrow 239.97 \text{ php}$

c) Pag-IBIG $\rightarrow 400 \text{ php}$

* Employee Share $\rightarrow 200 \text{ php}$

* Employer Share $\rightarrow 200 \text{ php}$

Other Computations

a) Basic Gross Pay = Daily Rate $\times$ Total Days Present

$= 460 \text{ php} \times 6 \text{ days} = 2,760 \text{ php}$

b) Overtime Pay = Hourly Rate $\times 1.25 \times$ OT Hours

$= 57.50 \text{ php} \times 1.25 \times 2 \text{ hrs} = 143.75 \text{ php}$

c) Regular Holiday Pay = Daily Rate $\times 2$ (200% double pay)

$= 460 \text{ php} \times 2 = 920 \text{ php}$

d) Special Holiday Premium : Daily Rate x 0.30 (30% premium added)

$$= 460 \text{ php} \times 0.30 = 138 \text{ php (added to basic pay for that day)}$$

$$= 138 \text{ php} + 460 \text{ php (basic pay)} = 598 \text{ php}$$

e) Late Deduction : Late Deduction Rate per minute x Minutes Late

$$= 1 \text{ php} \times 15 \text{ minutes} = 15 \text{ php deduction}$$

f) 13th Month Pay : Total Annual Basic Salary Earned $\div$ 12

$$= 71,990 \text{ php (pro-rated for 6 months)} \div 12 = 5,999.16 \text{ php}$$

